



Repeatable AI/ML Application Deployments on the Edge (RAADE)

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Problem Definition

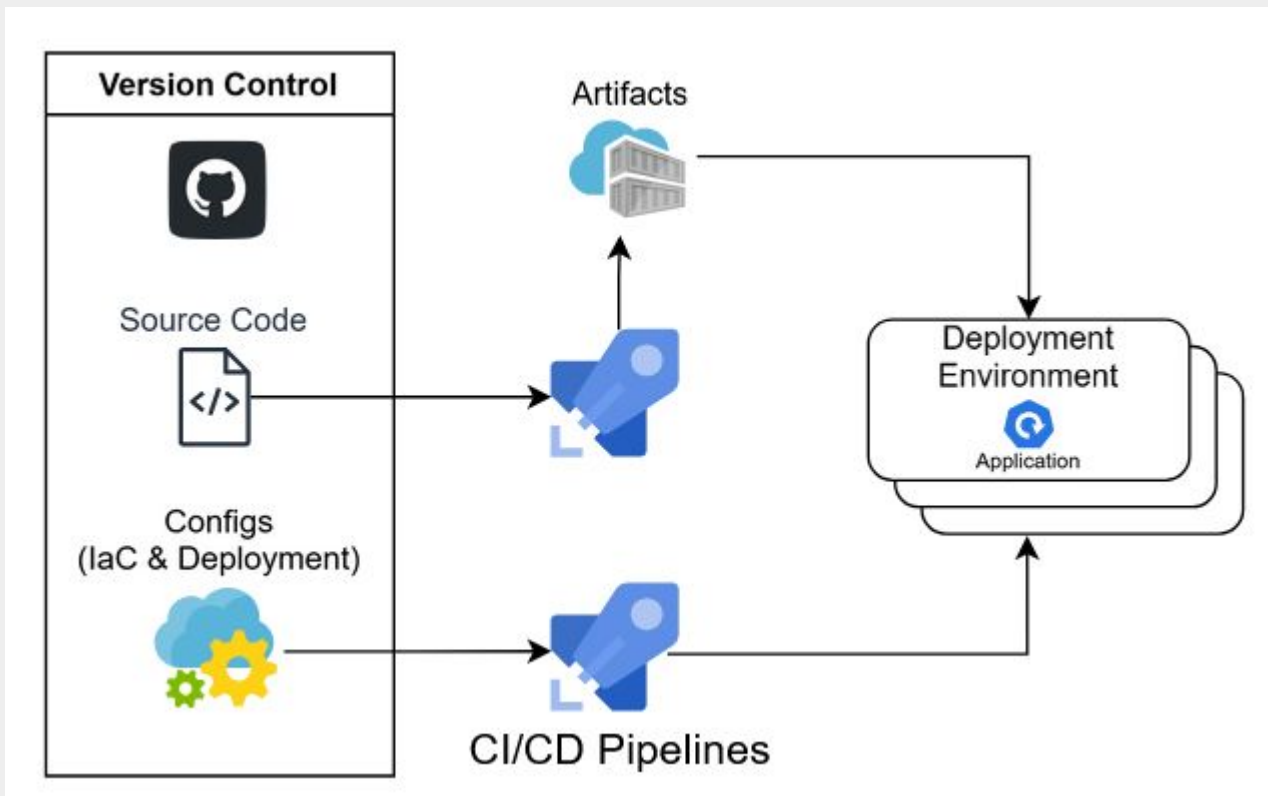
Deploying cloud-native applications onto the **Tactical Data Platform (TDP)**, an Army enterprise Platform-as-a-Service, via the AI2C's **Tactical Server Kit (TSK)** in **denied, disrupted, intermittent, and limited bandwidth (DDIL)** environments is currently a **manual and complex** process.

The **incompatibility of traditional DevSecOps tools** and processes in the environments **hinders** rapid, scalable, and **reliable application delivery** at the tactical edge and **prohibits frequent deployments** and rapid iteration.

This limits the AI2C's ability to rapidly integrate customer feedback and **prevents the delivery of new capabilities to the field** to generate timely value for the warfighter.

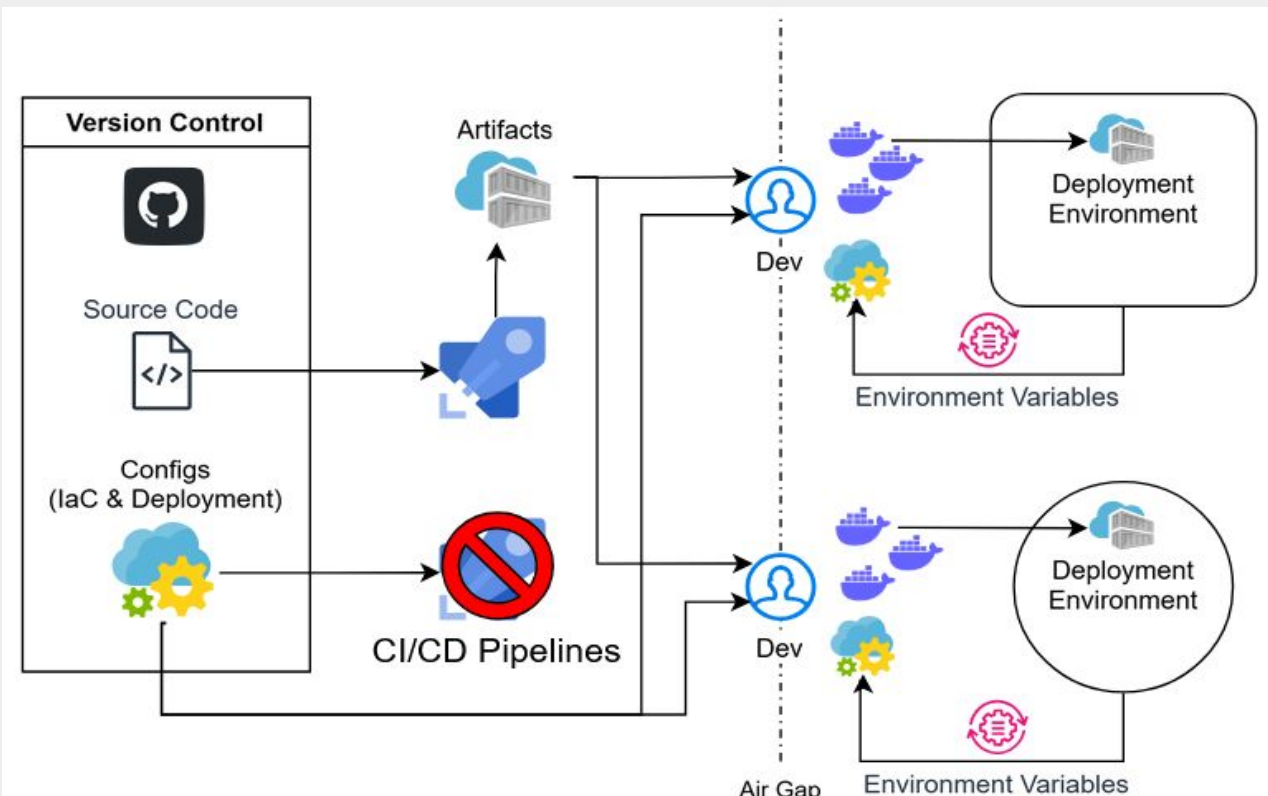
Traditional DevSecOps Deployment

- Version controlled source code
- Automated framework
- Automated deployment pipelines



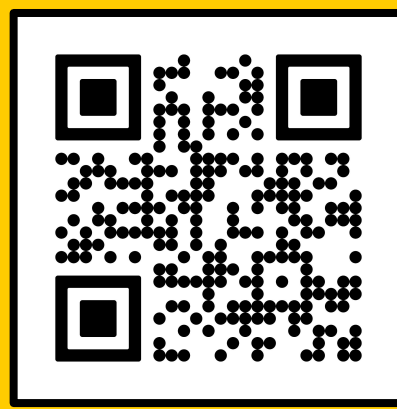
Disconnected Edge Deployment

- Locally controlled source code
- Manually controlled framework
- Non-existent deployment pipelines



Why ZARF?

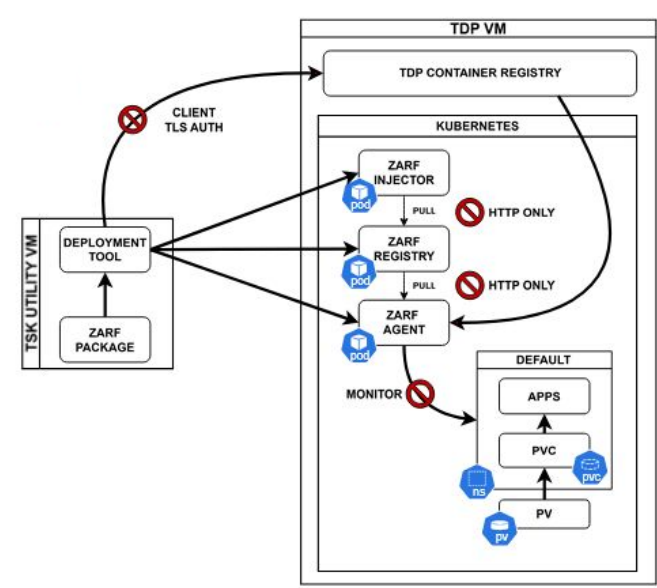
- Securely Package Apps & Resources
- Deploy Cloud Apps While Disconnected
- Easily Maintain Apps While Disconnected



Scan to learn more about ZARF!

RAADE Deployment and Testing using ZARF

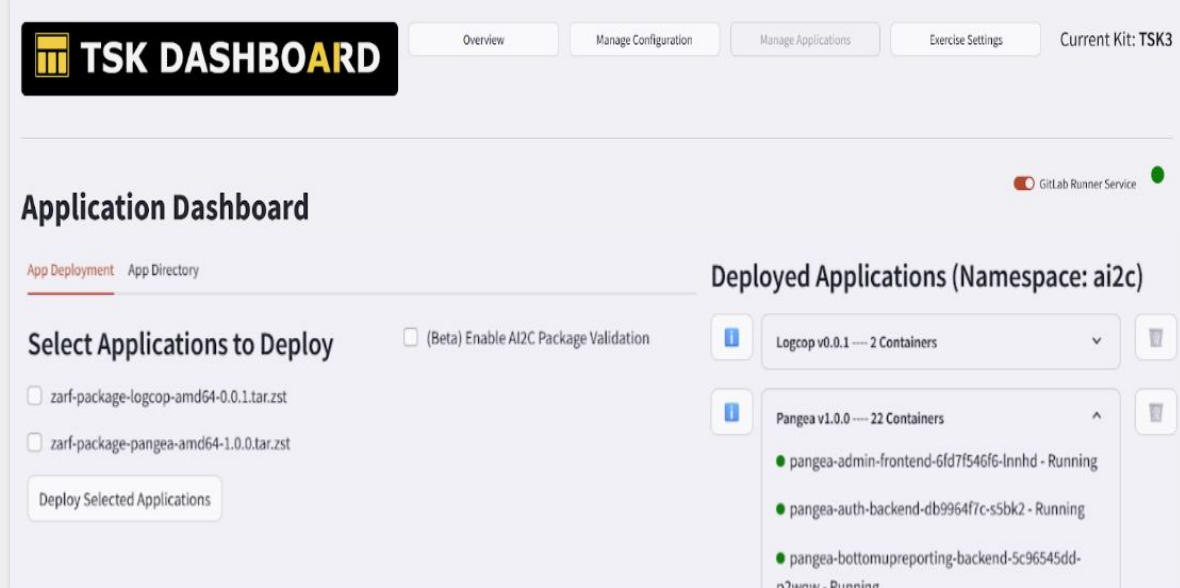
- Incompatible due to security constraints and architecture decisions
- Requires heavily modified Zarf init package



Identified compatibility issues between ZARF and TDP

As part of this capstone project, we focused on deploying **PANGAEA v2**, the most complex AI2C application available, consisting of **22 containerized microservices**. Traditionally, deploying this app to the **Tactical Data Platform (TDP)** required a deep understanding of the architecture, environment configurations, and manual steps—resulting in **significant time variability**.

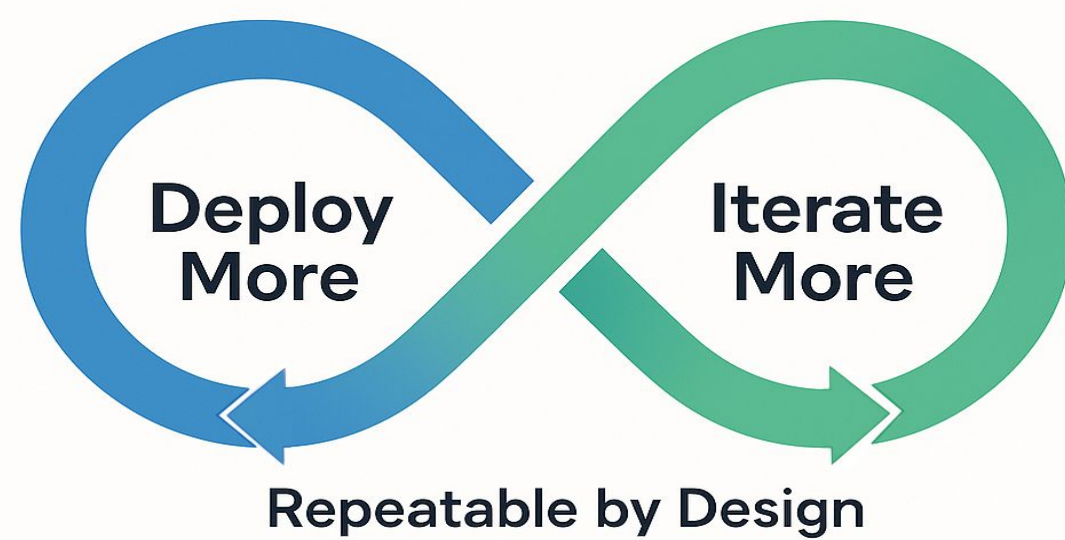
By using **ZARF** within the RAADE framework, we were able to package the entire deployment process, including dependencies and configuration logic, into a single portable artifact. This allowed us to **drastically reduce deployment time**, improve repeatability, and eliminate the need for prior domain knowledge.



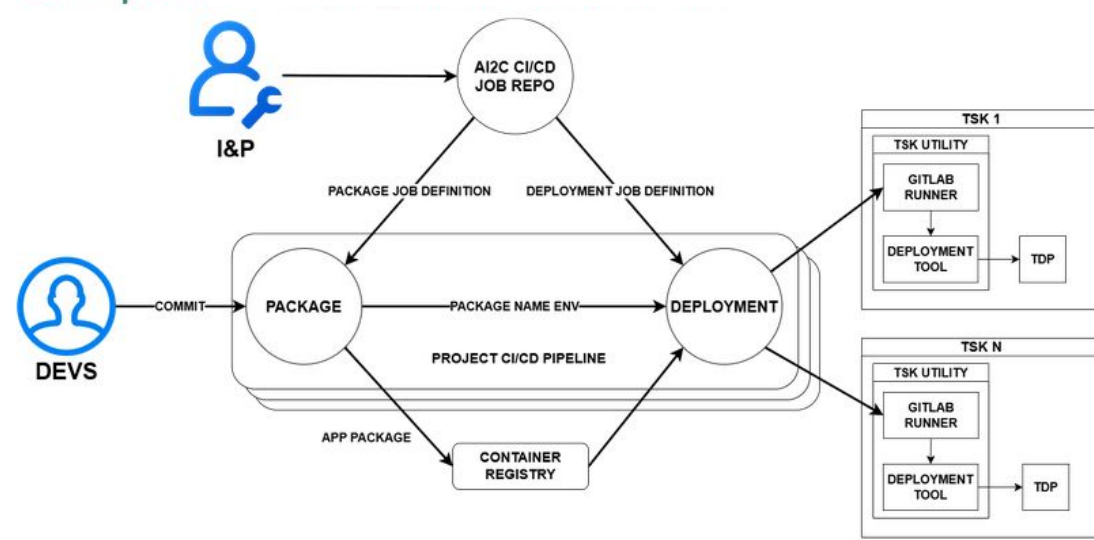
Easy-to-use dashboard for Application management and deployment

Deployment Time Comparison for PANGAEA v2

Deployment Method	Time Required	Notes
Current Deployment	120 – 960 min (2 – 16 hours)	- Highly variable - Dependent on operator expertise
RAADE Deployment with ZARF	~20 minutes	- Static - Requires no prior experience
Time Improvement	6 to 48 times faster	Consistent across repeated deployments



CD Pipelin



CD pipeline to deliver Application packages in a connected environment

Key Takeaways

- ZARF packaging removes complexity:** All artifacts, images, and scripts are pre-packaged—no manual setup required.
- Repeatable, scalable deployment:** Enables the same deployment regardless of who runs it.
- Reduced training burden:** Eliminates the need for specialized expertise in TDP deployment and application architecture.



Customized TDP ZARF Source Code and Container Images

JCWA R2D2 Account Required